

Saee Desai

sae9102@gmail.com | linkedin.com/in/saee-desai-237519206 | github.com/SaeeDesai | (901)-446-8602

SUMMARY

Applied ML Researcher specializing in NLP pipelines, LLM fine-tuning, and multimodal generative AI. Hands-on experience building transformer-based classification systems, diffusion model pipelines, and secure distributed data systems. Peer-reviewed publication in cryptographic security (Springer, 2025). Seeking entry-level ML Engineer or Data Scientist roles in the USA.

EDUCATION

University of Maryland, College Park

M.S. in Data Science, GPA: 3.85/4.0

Vishwakarma Institute of Information Technology

B.Tech in Computer Engineering, CGPA: 9.13/10

Maryland, USA

Aug 2024 – May 2026

Pune, India

Aug 2020 – May 2024

WORK EXPERIENCE

Applied Researcher

Do Quantum

Sep 2025 – May 2026

College Park, Maryland, USA

- Architected a **post-quantum secure healthcare data pipeline** integrating **ML-KEM-768**, **ML-DSA-65**, AES-256-GCM, and HKDF, enabling cryptographically verified data exchange across distributed hospital, gateway, and pharmaceutical systems.
- Built modular **Flask REST API** microservices simulating adversarial healthcare data exchange; implemented cryptographic validation workflows and packet-level audit logging for compliance traceability.
- Designed and executed adversarial robustness testing — simulating **replay, corruption, and ciphertext tampering attacks** — while benchmarking cryptographic pipeline throughput across healthcare file formats up to **10 MB**.
- Presented research findings at the **Do Quantum × MathQuantum Research Symposium** at the University of Maryland; contributed to technical reviews and iterative system refinement within a multidisciplinary research team.

TECHNICAL SKILLS

Languages: Python, SQL, C++

Machine Learning & Deep Learning: PyTorch, TensorFlow, scikit-learn, CNNs, Transformers, Diffusion Models, LoRA fine-tuning, CLIP, Transfer Learning

NLP & Generative AI: HuggingFace Transformers, LangChain, LangGraph, Large Language Models (LLMs), Text Classification, Emotion Analysis, Prompt Engineering

ML Systems & Deployment: AI Inference Pipelines, REST APIs, Flask, Model Serving, Model Evaluation, Confidence Scoring, Human-in-the-Loop (HITL) Systems

Data, Evaluation & Tools: Pandas, NumPy, Feature Engineering, Statistical Analysis, Cross-Validation, Data Augmentation, Git, Linux, Gradio, AWS S3, Docker (basic)

PROJECTS

Real-Time NLP Content Moderation and Emotion Analysis System | GitHub

Aug 2025 – Dec 2025

- Engineered an end-to-end **NLP pipeline** for real-time content moderation across Reddit posts and nested comment threads, integrating hate speech detection and 28-class emotion classification.
- Fine-tuned **DistilBERT** transformer models on **HateXplain** and **GoEmotions** datasets; achieved **76.25% hate detection accuracy** with optimized low-latency inference via model caching.
- Deployed a **Flask-based REST API** with modular inference architecture, confidence scoring, and thread-aware context analysis, designed for scalable production moderation workflows.
- Implemented **human-in-the-loop (HITL)** review interface enabling interpretable prediction auditing and feedback-driven model refinement.

Multimodal Emotion-to-Art Generation — Transformers and Diffusion Models | GitHub

Aug 2025 – Dec 2025

- Designed an **end-to-end generative AI system** mapping natural language emotion inputs to artwork using transformer-based emotion embeddings and latent diffusion models.
- Trained a **DistilBERT emotion classifier** across **27 classes**; mapped probabilistic emotion outputs to image generation controls, improving alignment between user intent and model output.
- Fine-tuned **Stable Diffusion with LoRA** under constrained GPU memory through mixed-precision training and batch size optimization, stabilizing convergence without performance degradation.
- Built a **CLIP-based evaluation pipeline** measuring semantic similarity between text prompts and generated images, identifying systematic failure modes in low-frequency emotion classes.

PUBLICATIONS

- **Enhanced Communication Security: Leveraging Quantum Cryptography** (*First Author*)

ICT Analysis and Applications (ICT4SD 2024), Lecture Notes in Networks and Systems, vol. 1197. Springer, March 2025.